

Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1 Let f be a function given by: $f(x) = (\tan^{-1}(x))^2$.

- (a) State the domain of f . Write your answer in interval notation.

Explanation. Domain of $f = (-\infty, +\infty)$

- (b) Find the critical points of f . Show your work!

Explanation. Since

$$f'(x) = 2 \tan^{-1}(x) \cdot \frac{1}{x^2 + 1} = \frac{2 \tan^{-1}(x)}{x^2 + 1},$$

the only critical points of f are the points where $f'(x) = 0$.

$$f'(x) = 0 \quad \equiv \quad \tan^{-1}(x) = 0 \quad \equiv \quad x = 0.$$

Therefore, the function f has the only critical point at $x = 0$.

- (c) Show that $f''(x) = \frac{2 - 4x \tan^{-1}(x)}{(x^2 + 1)^2}$.

$$\textbf{Explanation. } f''(x) = \left(\frac{2 \tan^{-1}(x)}{x^2 + 1} \right)' = \frac{2 \frac{x^2+1}{x^2+1} - 2 \tan^{-1}(x)(2x)}{(x^2 + 1)^2} = \frac{2 - 4x \tan^{-1}(x)}{(x^2 + 1)^2}$$

- (d) Use the **second derivative test** to determine whether critical point(s) of f correspond to local minimums, local maximums, or whether the test is inconclusive.

Problem 1

Explanation. Since

$$f''(0) = \frac{2 - 4(0) \tan^{-1}(0)}{(0^2 + 1)^2} = 2 > 0,$$

it follows by the Second derivative test that the function f has a **local minimum** at $x = 0$.

Problem 2

February 21, 2026

Problem 2 Determine the following information about the given function and then graph the function:

$$f(x) = \frac{x^2 + x + 1}{x^2}$$

Domain

x, y -intercepts

Symmetry

Asymptotes

Intervals of increasing/decreasing

Maxima/Minima

Intervals of concavity

Inflection points

Explanation.

(a) **Domain**

The function is a rational function, and so the domain of the function is all real numbers except where the denominator equals zero.

$$x^2 = 0 \implies x = 0$$

So the domain of f is $(-\infty, 0) \cup (0, \infty)$.

(b) **x, y -intercepts**

To find any x -intercept(s), set $y = 0$ and solve:

$$\begin{aligned} \frac{x^2 + x + 1}{x^2} = 0 &\implies x^2 + x + 1 = 0 \\ &\implies x = \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2(1)} \end{aligned}$$

which has no real solutions. Thus, f has no x -intercepts.

Since $x = 0$ is not in the domain of f , f has no y -intercepts as well.

(c) **Symmetry**

Note that $f(1) = 3$ and $f(-1) = 1$. So it cannot be for all values of x that either $f(-x) = f(x)$ or $f(-x) = -f(x)$. So f is neither even nor odd, and therefore f has no symmetry.

(d) **Asymptotes**

Vertical Asymptotes: Our only candidate is $x = 0$, and so we compute the two one-sided limits:

$$\lim_{x \rightarrow 0^-} \frac{x^2 + x + 1}{x^2} = \infty$$

$$\lim_{x \rightarrow 0^+} \frac{x^2 + x + 1}{x^2} = \infty$$

Therefore, $x = 0$ is the only vertical asymptote of f .

Horizontal Asymptotes: We compute the following limits:

$$\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x^2} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 + x + 1}{x^2} = 1$$

we checked both ends and so the only horizontal asymptote of f is $y = 1$.

(e) **Increasing/Decreasing**

$$\begin{aligned} f'(x) &= \frac{x^2(2x+1) - (x^2+x+1)(2x)}{x^4} \\ &= \frac{2x^3 + x^2 - 2x^3 - 2x^2 - 2x}{x^4} \\ &= \frac{-x^2 - 2x}{x^4} \\ &= \frac{-x - 2}{x^3} \end{aligned}$$

To find where f' is positive and where f' is negative, we need to find where $f'(x) = 0$ and where $f'(x)$ does not exist. Clearly, $f'(x)$ does not exist when $x = 0$. To find when $f'(x) = 0$, we solve:

$$\begin{aligned} \frac{-x-2}{x^3} = 0 &\implies -x-2 = 0 \\ &\implies -x = 2 \\ &\implies x = -2 \end{aligned}$$

Since $x = 0$ is not in the domain of f , $x = -2$ is the only critical point of f . To see where f is increasing and decreasing, consider the following sign chart for f' :

$$\begin{array}{ccccccc} & (-) & & (+) & & (-) & \\ & \text{blue} & & \text{red} & & \text{blue} & \\ \leftarrow & & | & & | & & \rightarrow f' \\ & & -2 & & 0 & & \\ f'(-3) = \frac{-1}{27} & & & f'(-1) = 1 & & f'(1) = -3 & \end{array}$$

Problem 2

So we see that f is increasing on $(-2, 0)$, and f is decreasing on $(-\infty, -2)$ and $(0, \infty)$.

(f) **Local Extrema**

f' changes from negative to positive at $x = -2$, so this is the location of a local minimum. f' also changes from positive to negative at $x = 0$, but f is not defined at $x = 0$ and so this is not a local extreme value. f has a local minimum at $\left(-2, \frac{3}{4}\right)$.

(g) **Concavity**

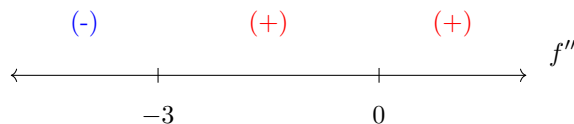
$$\begin{aligned} f''(x) &= \frac{x^3(-1) - (-x-2)(3x^2)}{x^6} \\ &= \frac{-x^3 + 3x^3 + 6x^2}{x^6} \\ &= \frac{2x^3 + 6x^2}{x^6} \\ &= \frac{2x + 6}{x^4} \\ &= \frac{2(x+3)}{x^4} \end{aligned}$$

To find where f'' is positive and where f'' is negative, we need to find where $f''(x) = 0$ and where $f''(x)$ does not exist. Clearly, $f''(x)$ does not exist when $x = 0$. To find when $f''(x) = 0$, we solve:

$$\begin{aligned} \frac{2(x+3)}{x^4} = 0 &\implies 2(x+3) = 0 \\ &\implies x = -3 \end{aligned}$$

The denominator of f'' is always positive so the sign of f'' depends on the numerator. When $x < -3$, $f'' < 0$ and when $x > -3$, $f'' > 0$.

To see where f is concave up and concave down, consider the following sign chart for f'' :



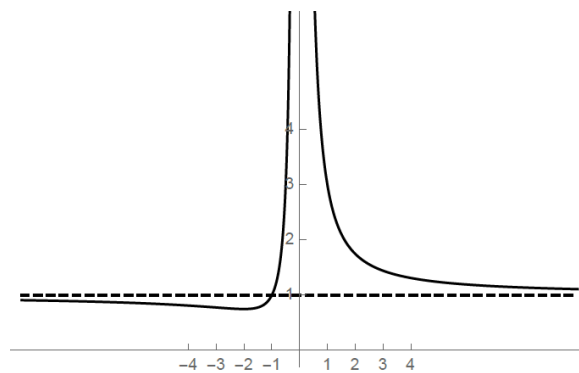
So we see that f is concave up on $(-3, 0)$ and $(0, \infty)$, and f is concave down on $(-\infty, -3)$.

(h) **Inflection Points**

$f''(x)$ changes sign from negative to positive at $x = -3$ so f has an inflection point at $\left(-3, \frac{7}{9}\right)$

Problem 2

(i) The graph of f



Problem 3

February 21, 2026

Problem 3 Given that:

$$\begin{array}{llll} \lim_{x \rightarrow -\infty} f(x) = 0 & \lim_{x \rightarrow -3^-} f(x) = \infty & \lim_{x \rightarrow -3^+} f(x) = -\infty & \lim_{x \rightarrow 4^-} f(x) = \infty \\ \lim_{x \rightarrow 4^+} f(x) = -\infty & f(1) = 1 & f(5) = -2 & \lim_{x \rightarrow 9} f(x) = 3 \\ f'(1) \neq 0 & f'(7) = 0 & f'(x) = 2 \text{ for } x > 9 & f''(1) = 0 \end{array}$$

the domain of f is: $(-\infty, -3) \cup (-3, 4) \cup (4, 9) \cup (9, \infty)$ and f is continuous on its domain. The following sign chart for the first and second derivatives of f :

	-3	1	4	5	7	9
$f'(x)$	+	+		+	-	+
$f''(x)$	+	-	+		-	0

find the following:

- Critical points.
- Intervals where f is increasing/decreasing.
- Local extrema.
- Inflection points.
- Intervals of concavity.
- Sketch the graph of f .

Explanation. (a) Critical points.

The critical points of f occur at points in the domain of f where either $f'(x) = 0$ or where $f'(x)$ does not exist. Even though $f'(-3)$, $f'(4)$, and $f'(9)$ do not exist, all three of those points are not in the domain of f and therefore are not critical points. We are given that $f'(7) = 0$, and so $x = 7$ is a critical point of f . Therefore, $x = 7$ is the only critical point of f .

- Intervals where f is increasing/decreasing.

f is increasing when $f'(x) > 0$. From the sign chart and our critical points, these are the intervals $(-\infty, -3)$, $(-3, 4)$, $(4, 7)$, and $(9, \infty)$. $f(x)$ is decreasing when $f'(x) < 0$. From the sign chart, this is on the interval $(7, 9)$.

- Local extrema.

Using the first derivative test, $f(7)$ is a local maximum because the derivative changes sign from positive to negative. So, $x = 7$ is the only local extremum of f , and it is a local maximum.

Problem 3

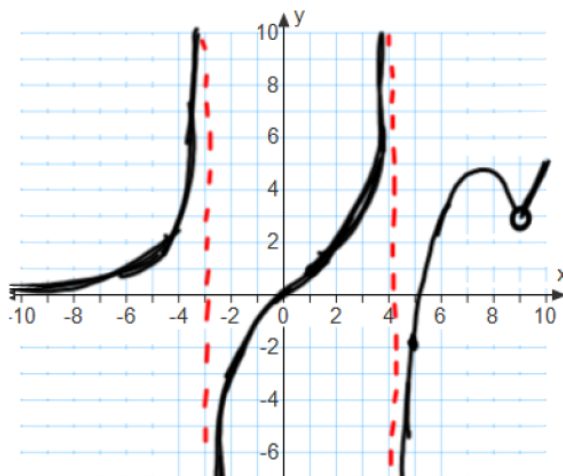
(d) Inflection points.

Possible inflection points occur where $f''(x) = 0$ or where $f''(x)$ does not exist. We are given that $f''(1) = 0$. In addition, $f''(x)$ does not exist at $x = -3, 4, 9$. However, these are not inflection points because f is not defined at these points. Since f'' changes sign at $x = 1$, this is in fact an inflection point of f . We are given that $f(1) = 1$, and so the only inflection point of f is the point $(1, 1)$.

(e) Intervals of concavity.

$f(x)$ is concave up when $f''(x) > 0$. From the sign chart, this is on the intervals $(-\infty, -3)$ and $(1, 4)$. $f(x)$ is concave down when $f''(x) < 0$. From the sign chart, this is on the interval $(-3, 1)$ and $(4, 9)$.

(f) Sketch the graph of f .



Problem 4

February 21, 2026

Problem 4 Determine the following information about the given function and then graph the function:

$$f(x) = 3 \sin \left(\frac{x}{2} \right), [-2\pi, 2\pi]$$

Domain

x, y -intercepts

Symmetry

Asympotes

Intervals of increasing/decreasing

Maxima/Minima

Inflection points

Intervals of concavity

Sketch a graph of f

Explanation. (a) Domain: This was given as $[-2\pi, 2\pi]$

(b) x, y -intercepts:

x -intercepts occur when $f(x) = 3 \sin \left(\frac{x}{2} \right) = 0 \implies \frac{x}{2} = 0 + k\pi$ where k is an integer, $-1 \leq k \leq 1$. On the domain this is $x = -2\pi, 0, 2\pi$.

y -intercepts: $f(0) = 3 \sin \left(\frac{0}{2} \right) = 0$ The y -intercept is the point $(0, 0)$

(c) Symmetry:

$f(-x) = 3 \sin \left(\frac{-x}{2} \right) = -3 \sin \left(\frac{x}{2} \right) \neq f(x)$ so f is not even $-f(x) = -3 \sin \left(\frac{x}{2} \right) = f(-x)$ so f is odd

(d) Intervals of increasing/decreasing:

$$f'(x) = 3 \cos \left(\frac{x}{2} \right) \cdot \frac{1}{2} = \frac{3}{2} \cos \left(\frac{x}{2} \right)$$

Since f' is defined everywhere on $(-2\pi, 2\pi)$ to find the critical points we just solve the equation $f'(x) = 0$ where $-2\pi < x < 2\pi$:

$$f'(x) = 0 \iff \frac{3}{2} \cos \left(\frac{x}{2} \right) = 0$$

$$\iff \cos \left(\frac{x}{2} \right) = 0$$

$$\iff -2\pi < x < 2\pi \text{ and } x/2 = \pi/2 + n\pi \text{ with } n \text{ an integer}$$

$$\iff -2\pi < x < 2\pi \text{ and } x = \pi + n2\pi \text{ with } n \text{ an integer}$$

$$\iff x = -\pi \text{ and } x = \pi.$$

So the only critical points are $x = -\pi$ and $x = \pi$.

If $-2\pi < x < -\pi$, it follows that $-\pi < x/2 < -\pi/2$ so $x/2$ lies in the third quadrant and $\cos(x/2) < 0$.

If $-\pi < x < \pi$, it follows that $-\pi/2 < x/2 < \pi/2$ so $x/2$ lies in the fourth and first quadrants and $\cos(x/2) > 0$.

If $\pi < x < 2\pi$, it follows that $\pi/2 < x/2 < \pi$ so $x/2$ lies in the second quadrant and $\cos(x/2) < 0$

$\implies f$ is increasing on the interval $(-\pi, \pi)$, and f is decreasing on the intervals $(-2\pi, -\pi)$ and $(\pi, 2\pi)$.

(e) Maxima/minima:

The first derivative test states we have a local maximum if the sign of f' changes from positive to negative, and a local minimum if the sign of f' changes from negative to positive.

Therefore there is a local maximum at $x = \pi$ and local minimum at $x = -\pi$ by the results in part (d).

(f) Intervals of concavity: Concavity can change at interior points in $[-2\pi, 2\pi]$ where f'' is undefined or equal to 0 are only *candidates* for the inflection points.

$$f''(x) = \frac{-3}{2} \sin\left(\frac{x}{2}\right) \cdot \frac{1}{2} = \frac{-3}{4} \sin\left(\frac{x}{2}\right)$$

$\frac{-3}{4} \sin\left(\frac{x}{2}\right) = 0$ when $\sin\left(\frac{x}{2}\right) = 0$ which in the domain of $-2\pi < x < 2\pi$ occurs at $x/2 = n\pi$ with n an integer. $-2\pi < x < 2\pi$ and $x = n2\pi$ with n an integer $\implies x = 0$

We have a candidate for an inflection point at $x = 0$.

If $-2\pi < x < 0$, it follows that $-\pi < x/2 < 0$ so $x/2$ lies in the third and fourth quadrant and $\sin(x/2) < 0$.

If $0 < x < 2\pi$, it follows that $0 < x/2 < \pi$ so $x/2$ lies in the first and second quadrant and $\sin(x/2) > 0$.

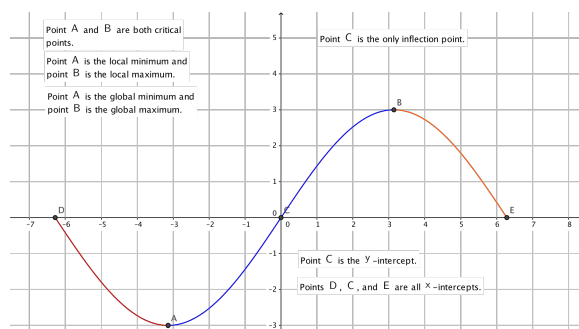
So f is concave up on the interval $(-2\pi, 0)$, and f is concave down on the interval $(0, 2\pi)$.

(g) Inflection points: Inflection points occur where the concavity of a function changes. Those interior points in $[-2\pi, 2\pi]$ where f'' is undefined or equal to 0 are only *candidates* for the inflection points. We found in part f that we had a candidate for an inflection point at $x = 0$. We verified in part f that concavity changes at $x = 0$.

Since $f(0) = 3 \sin\left(\frac{0}{2}\right) = 0$, the inflection point is $(0, 0)$

(h) Sketch the graph:

Problem 4



Problem 5

February 21, 2026

Problem 5 Determine the following information about the given function and then graph the function:

$$f(x) = x \ln(x)$$

Domain

x, y -intercepts

Symmetry

Intervals of increasing/decreasing

Maxima/Minima (including absolute)

Intervals of concavity

Inflection points

Given $\lim_{x \rightarrow 0^+} f(x) = 0$, sketch a graph f .

Explanation. (a) Domain: $(0, \infty)$

(b) x, y -intercepts:

x -intercepts occur when $f(x) = x \ln(x) = 0$. We would think this occurs when either $x = 0$ or $\ln(x) = 0$. However, $x = 0$ is not in the domain. We only have $\ln(x) = 0$ which is at $x = 1$. y -intercepts: None given the domain.

(c) Symmetry: Given the domain of the function, it does not make sense to check for even/odd.

(d) Intervals of increasing/decreasing:

$$f'(x) = \ln(x) + 1$$

$$\text{Find: } f'(x) = \ln(x) + 1 = 0 \implies \ln(x) = -1 \implies x = e^{-1}$$

Since f' possibly changes its sign at critical points only, the sign of f' is constant on intervals $(0, e^{-1})$ and (e^{-1}, ∞) . Since $f'(1) = 1 > 0$, and $f'(e^{-3}) = -3 + 1 = -2 < 0$, it follows that f' is negative on $(0, e^{-1})$ and positive on (e^{-1}, ∞) .

Therefore, f is decreasing on $(0, e^{-1})$ and f is increasing on (e^{-1}, ∞) .

(e) Maxima/Minima

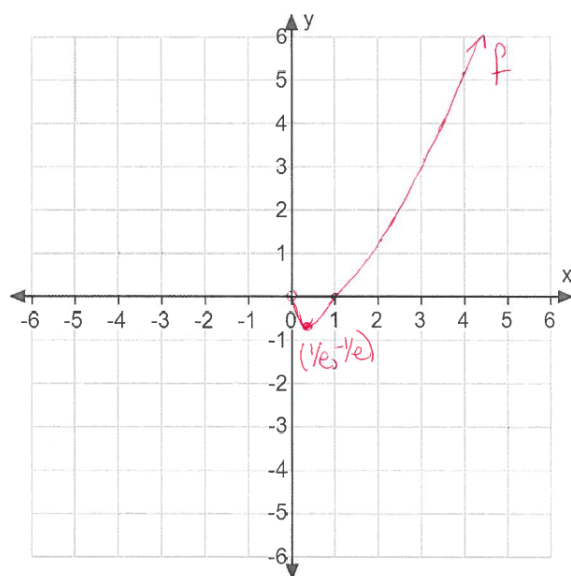
Since the function is decreasing on $(0, e^{-1})$ and f is increasing on (e^{-1}, ∞) , the absolute minimum value of f is attained at $x = e^{-1}$. The absolute minimum value of $f(x) = f(e^{-1}) = e^{-1} \ln e^{-1} = -e^{-1} = -\frac{1}{e}$

The absolute maximum value of f does not exist, since the only critical point is a minimum and $\lim_{x \rightarrow \infty} f(x) = +\infty$

(f) Intervals of concavity and inflection points: $f'(x) = \ln(x) + 1 \implies f''(x) = \frac{1}{x} > 0$ on the domain of f . Concave up on its domain and there are no inflection points.

Problem 5

- (g) Given $\lim_{x \rightarrow 0^+} f(x) = 0$, sketch a graph f :



Problem 6

February 21, 2026

Problem 6 Sketch a graph of the function f defined by

$$f(x) = \begin{cases} 2x - 2 \tan^{-1}(x) - x \ln(x^2 + 1) & \text{for } x \leq 1 \\ \sqrt{x-2} & \text{for } x \geq 2 \end{cases}$$

(a) Find the domain of f .

Explanation. The domain of f is $(-\infty, 1] \cup [2, \infty)$.

(b) Show that $f'(x) = -\ln(x^2 + 1)$ on the interval $(-\infty, 1)$.

Explanation. For $x < 1$ we have

$$\begin{aligned} f'(x) &= 2 - \frac{2}{1+x^2} - \ln(x^2 + 1) - x \cdot \frac{1}{x^2 + 1} \cdot 2x \\ &= 2 - \frac{2}{1+x^2} - \ln(x^2 + 1) - \frac{2x^2}{x^2 + 1} \\ &= 2 - \frac{2+2x^2}{x^2 + 1} - \ln(x^2 + 1) \\ &= 2 - 2 \cdot \frac{1+x^2}{x^2 + 1} - \ln(x^2 + 1) \\ &= 2 - 2 - \ln(x^2 + 1) = -\ln(x^2 + 1) \end{aligned}$$

(c) Compute a formula for f' on the interval $(2, \infty)$.

Explanation. For $x > 2$ we have

$$f'(x) = \frac{1}{2\sqrt{x-2}}$$

(d) Compute a formula for f'' on the interval $(-\infty, 1)$.

Explanation. For $x < 1$ we have

$$f''(x) = \frac{-1}{x^2 + 1} \cdot 2x = \frac{-2x}{x^2 + 1}$$

(e) Compute a formula for f'' on the interval $(2, \infty)$.

Explanation. For $x > 2$ we have

$$f''(x) = \frac{-1}{4(x-2)^{3/2}}$$

(f) List all the asymptotes (with limit justifications, if appropriate). (If f does not have a particular asymptote write “NONE”.)

(i) Vertical asymptotes:

Explanation. NONE: For $x \geq 2$, $f(x) = \sqrt{x-2}$ so no vertical asymptotes there. For $x \leq 1$, $f(x) = 2x - 2 \tan^{-1}(x) - x \ln(x^2 + 1)$ and we can look at each term. The term $2x$ is a polynomial. The term $-2 \tan^{-1}(x)$ is bounded. Finally, the term $-x \ln(x^2 + 1)$. We would be concerned about where $x^2 + 1 = 0$ but that never happens because of the squared term.

(ii) *Horizontal asymptotes:*

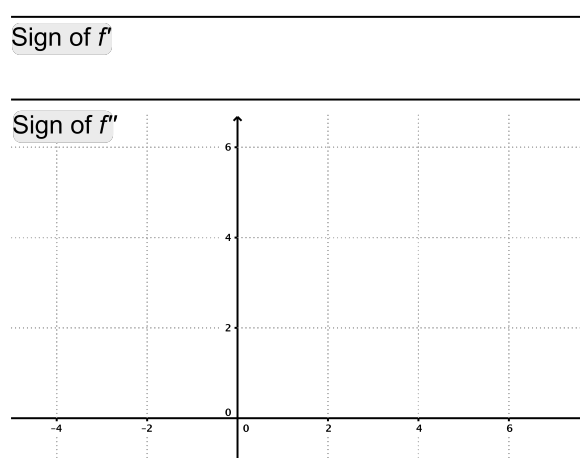
Explanation. No H.A. as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \underbrace{\sqrt{x-2}}_{\text{form } \infty} = \infty$$

No H.A. as $x \rightarrow -\infty$:

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} (2x - 2 \tan^{-1}(x) - x \ln(x^2 + 1)) \\ &= \lim_{x \rightarrow -\infty} [x \cdot (2 - \ln(x^2 + 1)) - 2 \tan^{-1}(x)] \\ &= \lim_{x \rightarrow -\infty} [x \cdot (2 + \ln(x^2 + 1)^{-1}) - 2 \tan^{-1}(x)] \\ &= \lim_{x \rightarrow -\infty} \left[\underbrace{x}_{\rightarrow -\infty} \cdot \left(2 + \ln \left(\underbrace{\frac{1}{x^2 + 1}}_{\rightarrow 0} \right) \right) - \underbrace{2 \tan^{-1}(x)}_{\rightarrow -\pi/2} \right] \\ &= \lim_{x \rightarrow -\infty} \left[\underbrace{x}_{\rightarrow -\infty} \cdot \left(2 + \underbrace{\ln \left(\frac{1}{x^2 + 1} \right)}_{\rightarrow -\infty} \right) - \underbrace{2 \tan^{-1}(x)}_{\rightarrow -\pi} \right] \\ &= \infty \end{aligned}$$

(g) In the following sign chart for f' , list the points (in the domain of f) where $f'(x) = 0$ or $f'(x)$ is undefined, indicate the sign of f' on each subinterval, and identify any local extrema (with “local max” or “local min”). In the following sign chart for f'' , list the points (in the domain of f) where $f''(x) = 0$ or $f''(x)$ is undefined, indicate the sign of f'' on each subinterval, and identify any inflection points. Use this information to sketch a graph of f in the blank grid. (Be sure to indicate any x - or y -intercepts.)



Explanation. Finding critical points of f :

For $x < 1$ we have

$$\begin{aligned} f'(x) = 0 &\iff -\ln(x^2 + 1) = 0 \\ &\iff \ln(x^2 + 1) = 0 \\ &\iff x^2 + 1 = e^0 = 1 \\ &\iff x = 0. \end{aligned}$$

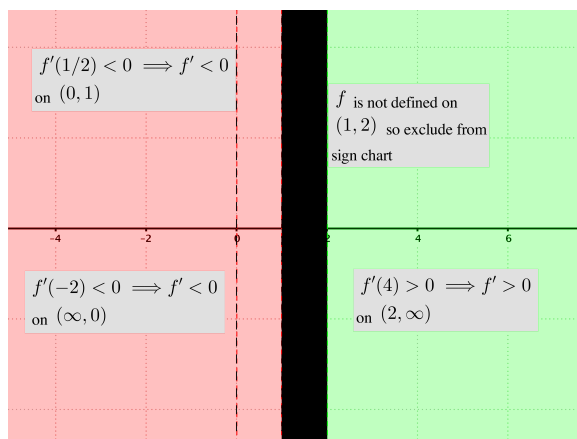
Critical point at 0 when $x < 1$.

For $x > 2$ we have

$$f'(x) = 0 \iff \frac{1}{2\sqrt{x-2}} = 0$$

This last equation is never true. Hence no critical points when $x > 2$.

Sign chart of f' :



Problem 6

Using the sign chart above, we see that the sign of f' does not change at $x = 0$ and is therefore neither a maximum or minimum. So f does not have a local maximum or local minimum.

Using the sign chart above, we see f' is positive on $(2, \infty)$ and thus intervals where f is increasing: $(2, \infty)$

Using the sign chart above, we see f' is negative on $(-\infty, 0)$, $(0, 1)$ and thus intervals where f is decreasing: $(-\infty, 0)$, $(0, 1)$.

Find candidates for inflection points:

For $x < 1$ we have

$$\begin{aligned} f''(x) = 0 &\iff \frac{-2x}{x^2 + 1} = 0 \\ &\iff x = 0. \end{aligned}$$

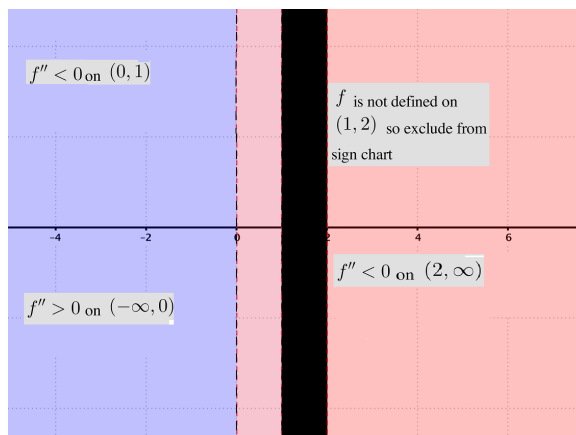
So, $x = 0$ is a candidate for an inflection point.

For $x > 2$ we have

$$f''(x) = 0 \iff \frac{-1}{4(x-2)^{3/2}} = 0$$

This last equation has no solution. Hence no candidates for inflection points when $x > 2$.

We need to determine the sign of f'' on the intervals $(-\infty, 0)$, $(0, 1)$, $(2, \infty)$.



f'' changes sign at $x = 0$ so $(0, 0)$ is an inflection point.

intervals where f is concave up: $(-\infty, 0)$

intervals where f is concave down: $(0, 1)$, $(2, \infty)$

Final sketch of graph:

Problem 6

